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DECLARATION

I hereby declare that this project report entitled "**IEEE Transactions on Systems, Man, and Cybernetics: Systems (2018)**" submitted by me for the award of the degree of **Bachelor of Technology** in the **Department of Artificial Intelligence**, Shri Vishnu Engineering College for Women (Autonomous), Bhimavaram , India, is a record of original project work carried out by me. This work has not been submitted either in part or whole to any other university or institution, in India or abroad, for the award of any degree.

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This is to certify that the Seminar Report entitled "**IEEE Transactions on Systems, Man, and Cybernetics:Systems (2018)**" that is being submitted by **V Yasaswi Lakshmi** [**22B01A45C3**], in partial fulfillment for the award of **Bachelor of Technology** in **Artificial Intelligence and Data Science** to Shri Vishnu Engineering College for Women (Autonomous), Bhimavaram is a record of bonafide work carried out by her under my supervision during the academic year 2025–26. To the best of my knowledge, this work has not been submitted elsewhere for the award of any degree.

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ABSTRACT

Blockchain technology is a decentralized distributed ledger system that enables secure, transparent, and tamper-resistant data storage without relying on a central authority. In recent years, blockchain has gained significant attention in healthcare due to increasing concerns regarding data security, interoperability, patient privacy, and transparency.

This seminar report presents a structured analysis of blockchain applications in healthcare based on a systematic review of 33 selected research publications conducted between 2008 and 2019

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. The review identifies major application areas such as data sharing, health record management, access control, audit trails, and supply chain management

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The findings indicate that although research in this domain is growing rapidly, most studies focus on theoretical frameworks and architectural designs rather than real-world implementations. Technical specifications such as consensus mechanisms, blockchain platforms, and smart contract usage are often not clearly defined

The study concludes that blockchain holds strong potential to transform healthcare systems, but practical deployment, scalability solutions, and regulatory standardization are necessary for real-world adoption.

ABBREVIATIONS

AI	Artificial Intelligence
EHR	Electronic Health Record
EMR	Electronic Medical Record
DNCASR	Discriminative Neural Clustering with Automatic Speech Recognition
PHR	Personal Health Record
POW	Proof of Work
POS	Proof of Stake
PBFT	Practical Byzantine Fault Tolerance
ICT	Information and Communication Technology
P2P	Peer-to-Peer
DER	Diarisation Error Rate

Table of Contents

DECLARATION **i**

ACKNOWLEDGEMENT **iii**

ABSTRACT **iv**

Abbreviations **v**

Table of Contents **vi**

List of Algorithms **vii**

1 Introduction **1**

1.1 Background 1

1.2 Problem Statement 1

1.3 Objectives 2

2 Literature Review **3**

2.1 Traditional Fire Detection Methods 3

2.1.1 Motion-Based Methods 4

2.1.1.1 Wavelet-Based and SVM Methods 4

2.2 Deep Learning Based Methods 4

3 PROPOSED METHODOLOGY **6**

3.1 Data Collection 6

3.2 Fire Detection Process 6

3.3 Fire Localization Algorithm 7

3.4 Severity and Object Detection 7

4 Results and Discussion **8**

4.1 Datasets Used 8

4.2 Results 9

5 Conclusions and Future Work **10**
5.1 Conclusions 10
5.2 Future Work 11

References **12**

CHAPTER 1

Introduction

1.1 Background

Fire is one of the most dangerous disasters that can cause severe damage to life, property, forests, and infrastructure. Traditional fire detection systems use smoke sensors, temperature sensors, or infrared detectors.

However, these systems:

Require close proximity, Cannot detect fire at early stages, Cannot provide location or severity information

With the development of smart CCTV surveillance systems, vision-based fire detection has become an important research area. Deep learning, especially Convolutional Neural Networks (CNNs), has shown powerful performance in image classification and object detection tasks.

This seminar focuses on a lightweight deep CNN model for efficient fire detection and localization in video surveillance systems.

1.2 Problem Statement

Existing fire detection methods have the following limitations:

- Color-based methods produce high false alarms
- Motion-based methods fail in fog/smoke environments

- High computational cost makes real-time implementation difficult

Therefore, there is a strong need for a secure, decentralized, and transparent framework to manage healthcare data efficiently.

1.3 Objectives

The objectives of this systematic review are:

- To develop a deep CNN-based fire detection system
- To reduce model size and computational complexity
- To perform fire localization (identify exact fire region)
- To determine fire severity and object under fire

CHAPTER 2

Literature Review

2.1 Traditional Fire Detection Methods

Early research in fire detection primarily relied on handcrafted features and rule-based systems. These traditional methods focused on identifying fire characteristics such as color, motion, and texture. Although these approaches laid the foundation for vision-based fire detection, they suffer from several limitations in real-world surveillance environments.

Color-Based Methods

Color-based fire detection techniques analyze pixel intensities using color models such as RGB, YCbCr, and HSI. Fire pixels generally exhibit high intensity in the red channel and specific chrominance properties in YCbCr and HSI color spaces. Researchers developed threshold-based rules to classify pixels as fire or non-fire based on these color characteristics.

However, these methods often misclassify fire-like objects such as red clothes, red vehicles, sunset light, and artificial lighting as fire. Since color information alone is insufficient to distinguish actual flames from similar-colored objects, these methods generate a high number of false positives. Additionally, changes in illumination, shadows, and camera quality further reduce their reliability.

2.1.1 Motion-Based Methods

Motion-based approaches detect fire by analyzing frame-to-frame variations in video sequences. Since flames flicker and exhibit dynamic motion, temporal differences between consecutive frames are used to identify potential fire regions.

Although motion features improve detection compared to pure color-based methods, they fail to distinguish between fire and other moving objects such as waving flags, moving vehicles, or human activity. Fog, smoke, and camera vibrations can also trigger false detections. Therefore, motion-based methods lack robustness in complex real-world surveillance scenarios.

2.1.1.1 Wavelet-Based and SVM Methods

To improve performance, researchers introduced texture and spatio-temporal feature extraction techniques such as wavelet transforms combined with classifiers like Support Vector Machines (SVM). These methods analyze the irregular patterns and energy variations in fire regions.

While these approaches provide better accuracy than simple color-based methods, they require manual feature engineering and parameter tuning. The computational cost is high due to feature extraction and classification stages. Moreover, performance heavily depends on selected thresholds and handcrafted features. These limitations make them unsuitable for large-scale real-time surveillance systems.

2.2 Deep Learning Based Methods

With the advancement of Artificial Intelligence and deep learning, Convolutional Neural Networks (CNNs) have significantly improved image classification and object detection tasks. Unlike traditional methods, deep learning models automatically learn hierarchical features directly from raw image data, eliminating the need

for manual feature engineering.

CNN-based approaches have shown superior performance in fire detection by learning complex visual patterns of flames. Models such as ImageNet pretrained networks provide strong feature extraction capabilities through transfer learning.

One of the widely used architectures is AlexNet, which demonstrated improved fire detection accuracy compared to handcrafted methods. However, AlexNet has a large model size (approximately 238 MB) and high computational complexity. This makes it unsuitable for embedded systems and resource-constrained surveillance cameras.

To overcome these limitations, lightweight CNN architectures such as SqueezeNet were introduced. These models reduce parameters and computational requirements while maintaining competitive accuracy. Therefore, the proposed system adopts a modified SqueezeNet-based CNN architecture to balance efficiency and detection performance.

CHAPTER 3

PROPOSED METHODOLOGY

3.1 Data Collection

The proposed system is designed to perform intelligent fire analysis in video surveillance environments. It integrates fire detection, localization, severity estimation, and object identification within a single framework.

The system performs the following major tasks: 1.Fire Detection – Identifies whether a video frame contains fire.

2.Fire Localization – Determines the exact region of fire in the frame.

3.Severity Estimation – Measures the burning degree based on fire pixels.

4.Object Identification – Identifies the object under fire (e.g., car, forest, building).

This integrated approach improves both detection accuracy and practical usability in real-world emergency scenarios.

3.2 Fire Detection Process

The fire detection process begins by extracting a frame from a video sequence. The frame is passed through the CNN model, which computes probability scores for two classes: Fire and Non-Fire.

The class with the highest probability is selected as the final prediction. If the frame is classified as Non-Fire, no further processing is performed. If classified as

Fire, the system proceeds to localization and severity analysis.

This two-stage approach ensures computational efficiency by performing additional processing only when necessary.

3.3 Fire Localization Algorithm

After fire detection, the system identifies fire regions using selected feature maps from intermediate CNN layers.

1.The localization process includes:

2.Extracting relevant feature maps

3.Computing mean activation values

4.Applying thresholding

Binarizing the feature maps

Segmenting fire regions The DNCASR system uses deep learning techniques for speaker-attributed speech recognition. Transformer-based models are selected due to their ability to process long speech sequences efficiently using attention mechanisms. Neural clustering methods are used to identify speakers based on embeddings.

3.4 Severity and Object Detection

Following segmentation, the number of fire pixels is calculated to estimate the burning severity. A higher number of fire pixels indicates a larger or more intense fire.

This semantic understanding enables emergency services to prioritize critical situations and respond effectively.

CHAPTER 4

Results and Discussion

4.1 Datasets Used

The performance of the DNCASR system is evaluated using both synthetic To evaluate the effectiveness of the proposed lightweight CNN-based fire detection system, experiments were conducted using two publicly available benchmark datasets: the Foggia Fire Dataset and the Chino Fire Dataset. These datasets are widely used in fire detection research because they contain diverse fire and non-fire scenarios captured in real-world environments.

The Foggia Fire Dataset consists of various indoor and outdoor fire videos, including both small-scale and large-scale fire incidents. It also includes challenging conditions such as fire-like objects, complex backgrounds, and varying lighting conditions. Similarly, the Chino Fire Dataset provides real-world fire scenarios recorded under different environmental settings. It contains video sequences captured in outdoor areas, including forest fires, building fires, and open-area fire accidents.

Both datasets include difficult cases such as fog, sunlight reflections, red-colored objects, and smoke interference. These scenarios are critical for testing the robustness of fire detection systems because traditional methods often fail under such conditions. By evaluating the proposed model on these diverse datasets, the system's performance under realistic surveillance conditions is thoroughly validated. This ensures that the developed model is not limited to controlled labo-

ratory environments but is capable of functioning effectively in real-world CCTV applications.

4.2 Results

Experimental evaluation shows that the proposed CNN-based fire detection model achieves an overall accuracy of approximately 94.5

In addition to classification accuracy, the system demonstrates real-time processing capability. When deployed on a high-end GPU system, the model processes video frames efficiently without noticeable delay. Furthermore, even when deployed on embedded platforms, the system maintains acceptable performance.

Compared to traditional handcrafted feature-based methods, the proposed deep learning model provides several improvements. It offers higher detection accuracy, significantly reduces false alarms, and requires a much smaller model size. Additionally, the computational cost is lower due to the optimized architecture, making it suitable for practical surveillance applications.

CHAPTER 5

Conclusions and Future Work

5.1 Conclusions

This seminar presented a lightweight deep CNN-based fire detection system designed for real-time surveillance applications. The proposed system effectively addresses the limitations of traditional fire detection methods by automatically learning discriminative features from image data.

The system successfully detects fire at early stages, reduces false alarms, and accurately localizes fire regions within video frames. In addition to detection, it estimates fire severity and identifies the object under fire, thereby enhancing the system's practical usefulness for emergency response services.

By reducing model size and computational complexity, the proposed architecture achieves an optimal balance between accuracy, efficiency, and cost. Experimental results confirm that the system performs reliably in both indoor and outdoor environments, even under challenging conditions such as fog and sunlight reflections.

Therefore, the proposed system represents a practical and scalable solution for intelligent fire surveillance systems.

5.2 Future Work

Although the proposed system achieves promising results, several improvements can be considered for future research. One important direction is enhancing detection performance under extreme weather conditions such as heavy rain, dense smoke, and low-light environments.

Further optimization can be performed to reduce false positive rates and improve detection precision. The integration of advanced edge-AI hardware can enable the development of fully autonomous smart surveillance cameras.

Additionally, extending the system to include smoke detection and multi-hazard monitoring can further enhance early warning capabilities and improve overall fire safety management.

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